EGE KAYA

Vancouver, BC | Email | LinkedIn | Google Scholar | Website | GitHub

Profile

M.Sc. student in Computer Science at UBC (NLP and human-centred AI), working in the UBC HCI-NLP Lab and the Nunez Lab for Computational Mental Health and Cancer Care, with a B.Sc. in Computer Engineering from Politecnico di Torino. Work spans NLP interpretability, analysis and control research, health-AI study design with BC Cancer, and low-level systems engineering, including a Google Summer of Code project on the Swift Concurrency runtime.

Interests: LLM analysis, interpretability and evaluation (hidden-state probing, activation patching, subliminal learning); retrieval-augmented generation; explainable AI; human–AI and human-computer interaction.

Education

University of British Columbia **Vancouver, BC**
M.Sc. in Computer Science (thesis-based) *Sep. 2026 – Jun. 2028*

- Supervised by Prof. Cristina Conati; also working with Dr. John-Jose Nunez. Focus: explainable and human-centred AI, LLM analysis, control, and interpretability, Reinforcement Learning, AI Evaluation.
- Coursework: Analysis of NLP Models; Open Questions in NLP; Machine Learning Theory; AI Evaluation; Reinforcement Learning; Human-Computer Interaction.

Politecnico di Torino **Turin, Italy**
B.Sc. in Computer Engineering *Sep. 2023 – Sep. 2026*

- GPA 3.7/4.0 (27.5/30 average); final mark 101/110. Full merit scholarship.
- Selected coursework: Computer Architecture (30L/30); Cybersecurity and National Defence (30L/30); Object-Oriented Programming (29/30); Linear Algebra and Geometry (29/30); Algorithms and Data Structures; Operating Systems; Digital Electronics Design; Computer Networks; Data Science and Visualization; Introduction to Databases.

Experience

UBC and BC Cancer **Vancouver, BC**
Graduate Research Assistant, Human-Centred AI and NLP *Sep. 2026 – Present*

- Contribute to an LLM- and RAG-based cancer-care navigation assistant that connects users with oncology and supportive-care information.
- Study how recommendations should communicate evidence and explanations: why a recommendation was made, how it is grounded, and when to suggest further resources in sensitive situations.
- Helped design a two-hour stakeholder workshop for the BC Cancer Summit (19 Nov. 2026; 50–100 expected participants) combining individual chatbot interaction (simple, complex, and emotionally difficult scenarios) with round-table discussion among patients, caregivers, clinicians, administrators, and researchers.
- Contributed to the explanation-preference widget and study materials; the widget logs user preferences separately from chatbot responses.

Google Summer of Code – Swift **Remote**
Open-Source Contributor, Swift Concurrency Runtime *May 2026 – Sep. 2026*

- Implemented a 64-shard global task registry in the Swift Concurrency runtime so suspended async tasks are discoverable by debuggers and crash-analysis tools.
- Designed concurrent C++ data structures and synchronization paths, keeping measured overhead below 5% on the Swift benchmark suite.
- Extended LLDB Python tooling for live task enumeration and contributed to signal-safe crash-backtrace integration.

Politecnico di Torino **Turin, Italy**
Student Machine Learning Engineer and Researcher *Feb. 2026 – Jul. 2026*

- Developed models predicting student academic trajectories from institutional data: data preparation, feature engineering, cohort analysis, and evaluation.
- Built data-processing and evaluation pipelines for large tabular datasets; supported integration of AI-based class-support tools into university infrastructure.

Parma Calcio 1913 **Parma, Italy**
Software Engineer Intern *Sep. 2025 – Feb. 2026*

- Built production React/TypeScript applications for medical-staff workflow management and operational tooling.
- Implemented AWS S3/CDN asset delivery and real-time data pipelines, cutting load times by 80% and reaching Lighthouse scores above 99.

PoliTO Rocket Team **Turin, Italy**
Embedded Software Engineer *Feb. 2025 – Feb. 2026*

- Wrote STM32 flight software in C/C++: peripheral drivers, watchdogs, fault handling, and SWD debugging under real-time constraints.
- Built internal tools used by 150+ team members, reducing administrative overhead by 95%.

Research

LLM Forgetting: Hidden-State Probing and Activation Patching *2026 – Present*
In progress, with Finn Vamosi. Studying how language models behave when instructed to forget information, using hidden-state probing and activation patching to test how interventions at selected activations relate to later behaviour.

Subliminal Learning from Weight-Finetuned Teachers *2026 – Present*
In progress, with Finn Vamosi, Erell Boutin and Zung Nguyen. Studying whether behavioural traits of a fine-tuned teacher model can emerge in a student, and how weight updates and LoRA configurations relate to that transfer.

Explainable AI for Cancer Care Navigation *2026 – Present*
Manuscript in preparation (UBC–BC Cancer). Study of explanation needs for cancer-navigation recommendations, informed by a 16-paper review of healthcare XAI, trust, empathy, and personalised explanations, plus workshop participant data from the BC Cancer Summit 2026. Compares why/how/no-explanation options, text/visual/conversational formats, and direct versus supportive-resource recommendations.

Analysis of AI Usage in University-Level Lecturing *2026*
Public report, non-peer-reviewed. With A. Kale and B. Demirtola. Evaluated AI systems for lecturing and instructional-material generation, including a Computer Architecture lecture.

AI-Based Anomaly Detection in Network Security *2026*
Course report, non-peer-reviewed. With A. Kale and D. E. Bulte. Investigated ML approaches to anomalous activity in network-security data.

Selected Projects

Gravitational Lens Detection (ML4SCI / DeepLense) *Python, PyTorch*

- Trained ResNet-18 models for lens detection under 100:1 class imbalance: 0.9877 AUC, 96.9% sensitivity.
- Reached 0.9919 AUC on gravitational substructure classification, above the published evaluation baseline.

GitSimulator (DAG-Based Git Simulation) *Python, CI/CD, PyPI*

- Built a DAG simulation engine for merge, rebase, reset, and cherry-pick with live commit-graph visualisation.
- Published as an open-source PyPI package with 135+ automated integration tests.

Skills

ML and Research: Python, PyTorch, scikit-learn, NLP, LLM evaluation, probing and activation patching, RAG, explainable AI, computer vision.
Languages: Python, C, C++, Java, TypeScript, SQL, Swift.
Systems and Tools: Linux, LLDB, STM32, React, Next.js, Node.js, PostgreSQL, AWS, Docker, Git, CI/CD.

Honours

- **Top 0.34%** nationwide, Turkish University Entrance Examination (about 1.5 million candidates).
- **Top 9 finalist**, Amazon Robotics Hackathon (graph-routing algorithm for warehouse pods).
- **3rd place**, GitKraken Game Jam (GitSimulator).
- **KU Leuven CS Scholarship**: selected among the top ten incoming students (offer not taken up).
- **Top 10 graduate**, KAL, 2023.